

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question
Paper 3

20 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	72

Answer **all** the questions.

- 1 In a maternity ward at a local hospital, a mix-up involving three couples and three babies caused a lot of confusion. Based on phenotypic characteristics, the nurses were unable to correctly identify the parents of the babies. In order to solve the case, a scientist was called in to carry out a DNA test to identify the parents of the babies. The test was based on the principle that different individuals have a different number of repeating units at a particular locus in a chromosome.

(a) State the name of the test that was carried out. [1]

Paternity Test/ DNA Fingerprinting/Genetic Fingerprinting

Chromosome 13 was isolated from the DNA samples that were obtained from the three couples and three babies and used for further analysis. The sequence below shows a segment of chromosome 13, which was used in the analysis where (TTAGGAT) is the repeating unit and n is the number of repeats.

5' ...GCTAAGTATTGCTCAAGA... (TTAGGAT) $_n$...GATAAATAACTGGCTAGTA...-3'
3' ...CGATTTCATAACGAGTTCT... (AATCCTA) $_n$... CTATTTATTGACCGATCAT...-5'

The gel diagram in Fig. 1.1 shows the results of the DNA test obtained from each individual, Dad and Mom #1, #2 and #3, and baby A, B, and C.

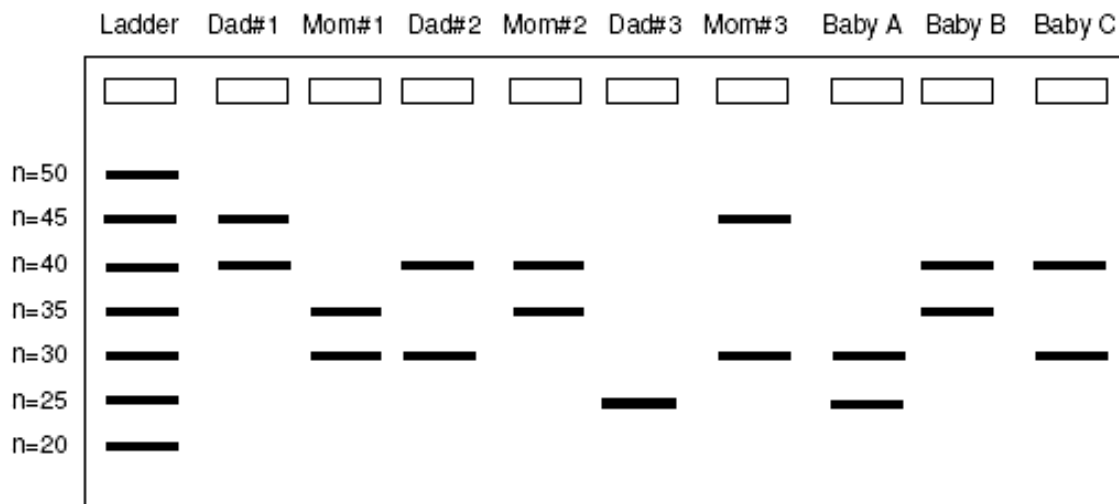


Fig. 1.1

(b) (i) Explain the purpose of the DNA ladder in Fig. 1.1. [1]

Each band serves as a comparison / reference for the size / no. of repeats present / found in the individuals tested.

(ii) DNA is colourless.

State how the DNA bands in the gel can be made visible. [1]

- Ethidium bromide under UV light
- staining with methylene blue
- autoradiography with a labeled probe

(c) With reference to Fig. 1.1, explain why

(i) some bands in a lane are higher than others; [2]

- Higher bands are longer as the DNA fragments have a larger number of repeated units

- The longer a DNA fragment, the harder it is for that fragment to travel through the pores in the gel, and so it migrates slower than the smaller fragments with fewer repeats.

(ii) the lane for Dad#3 has only one band; [2]

- They are homozygous at this locus.
- they have the same number of repeats on both homologous chromosome13
- For example: Dad #3 has 25 repeats on both versions of his chromosome 13.

(iii) some lanes have two bands. [2]

- They are heterozygous at this locus.
- they have a different number of repeats on each homologue of chromosome 13.

(d) (i) With reference to Fig. 1.1, state and explain which couple is the parents of Baby A. [2]

- couple #3
- Baby A has a band in n=25 and Dad #3 is the only parent with n=25 @ chromosome.

(ii) Explain why results of paternity testing are not absolutely certain. [2]

- Only a limited number of loci were considered.
- It is possible that a male may be a match at all the loci considered, but not be a match at another loci that was not considered in the test.

[Total: 13]

- 2 One approach of gene therapy to treat cystic fibrosis uses viruses to deliver normal alleles of the CFTR gene into epithelial cells of the airways.

A team of researchers in the USA developed a new strain of non-pathogenic adeno-associated virus (AAV), AAV2.5T. Genes for the CFTR protein and the enzyme luciferase were inserted into the DNA of the viruses. Luciferase catalyses the production of a green fluorescent protein when luciferin is added.

The normal AAV strain and the AAV2.5T strain were added to cultures of epithelial cells from the airways. After adding luciferin, the number of cells that had taken up the viral genes was estimated using the intensity of the green fluorescence which developed. The results are shown in Fig. 2.1.

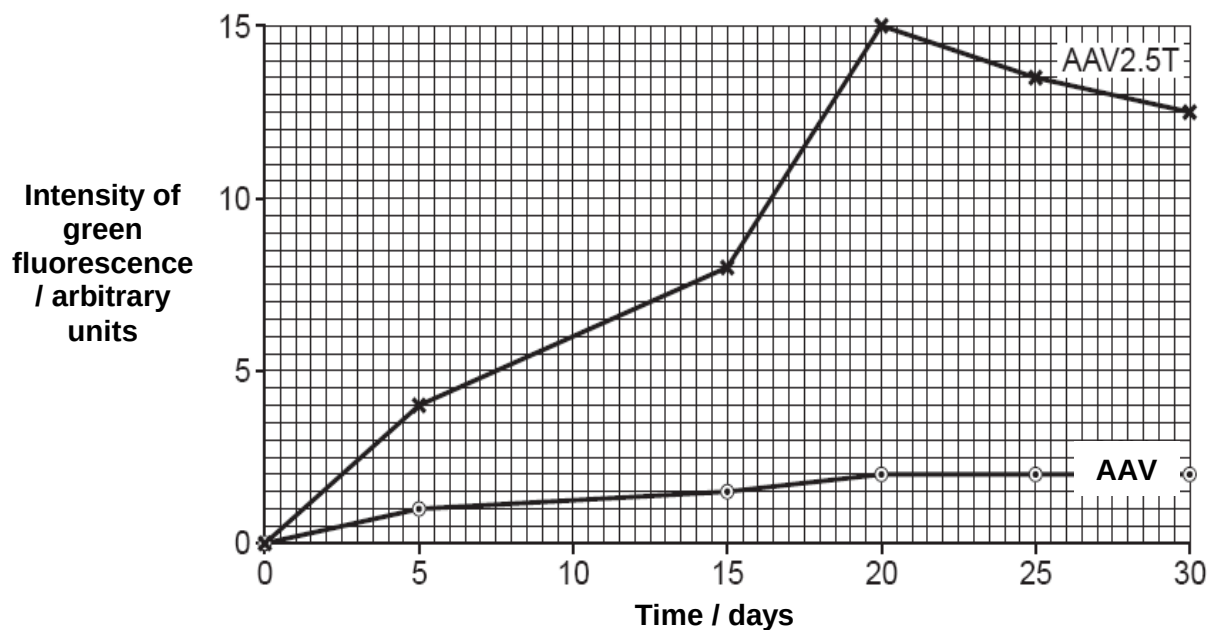


Fig. 2.1

- (a) Explain why the researchers inserted a gene for luciferase into the viral DNA. [1]
- **infected cells are able to produce luciferase;**
 - **able to identify infected cells/cells that have taken up viral DNA (which will fluoresce when luciferin is added)/ ref to gene for luciferase functions as a reporter gene;**
- (b) With reference to Fig. 2.1, compare the ability of the two viral strains, AAV and AAV2.5T, to infect epithelial cells of the airways. [2]
- Both AAV and AAV2.5T can infect epithelial cells;**
[Both pt 2 & 3 - 1m]
 - AAV2.5T infects more cells than AAV / infects cells more readily;**
 - Quote relevant values (both intensity and day indicated) from graph;**
[Both pt 4 & 5 - 1m]
 - Intensity of green fluorescence increases more in AAV2.5T compared to AAV from 0-20 days;**
 - from 0 to 15 au in AAV2.5T compared to only 2 au in AAV; A! any values correctly comparing rate of increase anytime during the first 20 days period**
[Both pt 6 & 7 - 1m]
 - From the 20th to 30th day, ability of AAV2.5T to infect cells fall but AAV's ability remains constant;**
 - Intensity of green fluorescence decreases to 12.5 au in AAV2.5T compared to remaining at 2 au in AAV; @ ½m**

- (c) Suggest why a decrease in intensity of green fluorescence was detected in cells infected with AAV2.5T during the last 10 days. [2]
- **Green fluorescent protein was broken down;**
 - **Luciferin was used up;**
 - **(Infected) cells die (as virus regain virulence);**
- R! cells constantly being shed/killed by immune system as gene therapy was done in cultures of epithelial cells.**
- (d) Describe how delivering normal alleles of the CFTR gene into epithelial cells in the airways could relieve the symptoms of cystic fibrosis. [4]
- 1m, max 3**
- **normal allele of CFTR gene is transcribed and translated / expressed to produce functional CFTR protein;**
 - **CFTR protein embedded in cell membrane allows chloride ions to leave cell / serves as functional chloride channel;**
 - **water also leaves cell down the water potential gradient;**
 - **normal / less viscous mucus formed ;**
- (essential point)**
- **no blockage of airways / less chance of bacterial lung infections/any other related possible symptoms;**
- (e) Researchers have started to use cDNA of the CFTR gene for more effective therapeutic results. Explain why cDNA of the normal CFTR gene is used instead of the normal CFTR gene. [2]
- cDNA does not contain non-coding regions such as introns;**
cDNA can be transcribed and translated directly without the need for post-transcriptional modifications such as splicing;
Hence the CFTR proteins can be synthesized efficiently;
- (f) AAV does not cause human disease, but because we are routinely exposed to this virus, 30 to 60 percent of people develop antibodies that neutralize AAV if it enters the circulation.

To extend the potential benefits of gene therapy to a broader population, researchers have produced a bioengineered decoy, which are empty AAV capsids disabled of their ability to enter target cells.

Suggest how using both the decoy and vector at the same time would improve the success rate of the gene therapy. [3]

Higher ratio of decoy to vector;
Decoy cannot enter target cells, thus they stay in circulation for longer period of time;
Ref to antibodies bound to the capsid decoys;
Allowing the DNA-carrying vectors to evade the antibodies and enter the targeted cells in the liver;

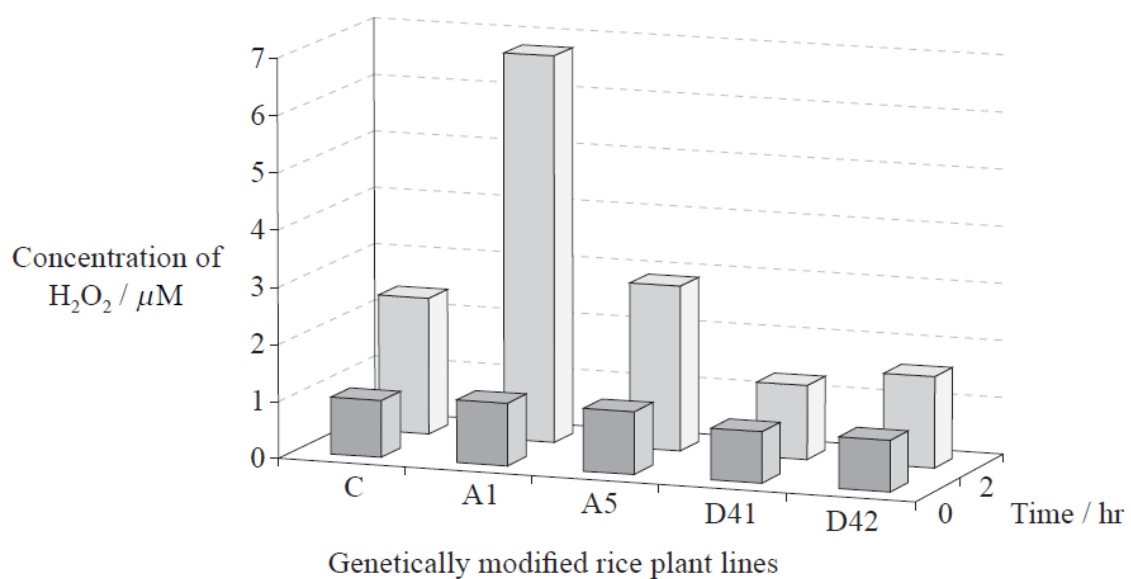
[Total: 14]

3

Plants have developed defense mechanisms against pathogens such as bacteria, fungi, and viruses. Chemicals released by these pathogens can trigger a defense response in infected plant cells. For example, the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and cellular chemicals, eventually kills both the cell and the pathogen.

The *OSRac1* gene was isolated and introduced into a number of rice plant (*Oryza spp.*) lines to study its role in disease resistance of plants to Blast fungus. Experiments were carried out to see if the *OSRac1* gene was part of the signalling pathway for hydrogen peroxide production.

A control and four other genetically modified rice plant lines were exposed to chemicals known to initiate a defense response and the production of hydrogen peroxide. The results are shown in Fig. 3.1 below.



Key: C: control

A1 and A5: rice plants with the *OSRac1* gene always turned on

D41 and D42: rice plants with the *OSRac1* gene suppressed

Fig. 3.1

- (a) With reference to Fig. 3.1, compare the changes in H_2O_2 production in the control and genetically modified plants after the chemical was applied and conclude if *OSRac1* gene is involved in disease resistance. [4]

- Production of H_2O_2 increased less in C (control) plants than A1/A5 plants supported with relevant data;
- Production of H_2O_2 in D41/D42 plants increased less than in C (control) plants supported with relevant data;
- Production of H_2O_2 increased the most in A1/least in D41 plants supported with relevant data;

[Compulsory conclusion point]

- Transgenic plants with functioning gene showed more H_2O_2 production so hypothesis; OR
- Transgenic plants with gene suppressed showed less H_2O_2 production so hypothesis;

(b) Suggest **two** possible concerns about the use of transgenic plants with the disease resistance gene.

- Disease resistance could spread to species related to rice, e.g. weeds;
 - Reluctance of consumers to eat genetically modified food products;
 - Outcompete natural species and change gene pool;
 - May increase allergenic effects in human/cattle;
- Any two above

[2]

(c) Edible vaccine against cholera using rice was developed by a team of Japanese scientists by cloning a gene from cholera bacteria, *Vibrio cholerae* into the genome of the Kitaake rice plant. The gene expressed a subunit of the disease-causing cholera toxin B. The steps are shown in Fig. 3.2.

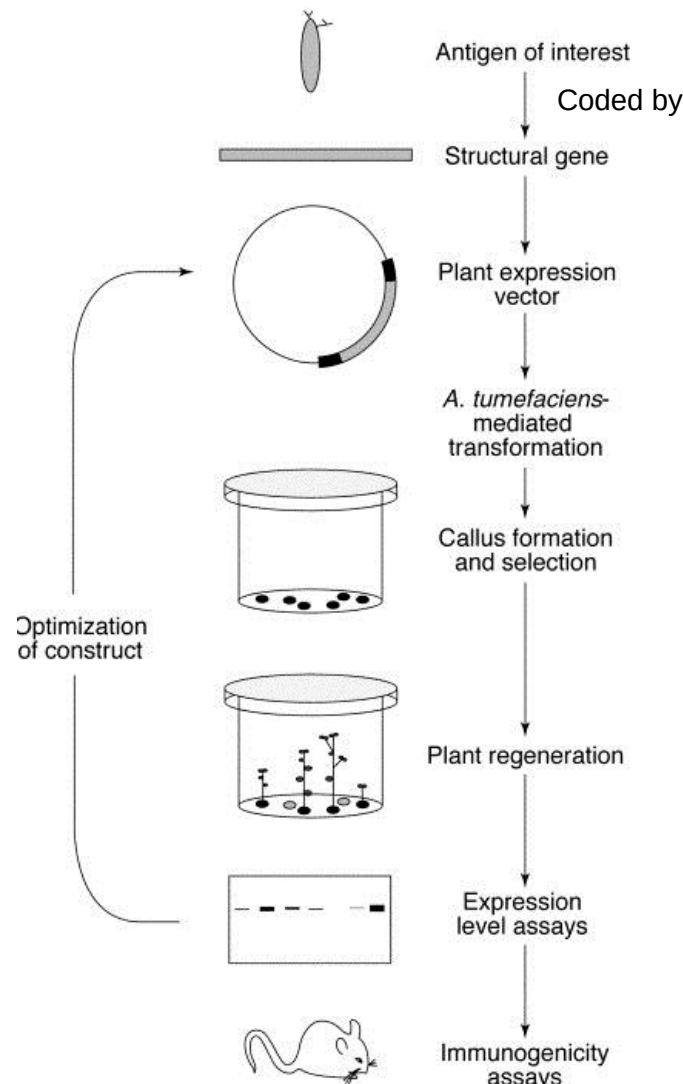


Fig. 3.2

(i) Describe how the antigen gene inserted into rice DNA genome would result in successful vaccination once the rice is eaten. [2]

- **[Essential compulsory point]** Antigen protein would be expressed in rice cells;
- Any 1 below:
- Antigen would trigger immune system to produce the antibody;
 - Antibody have a specific configuration that is complementary to the glycoprotein (antigen);
 - Antibody will be synthesized to mount an immune response against a subsequent vaccination against cholera bacteria;

Extend Q – N06/P2/Q8c - Outline the roles of glycoproteins in the cell surface membrane. [8]

- Aid cell recognition
- Act as cell surface receptors (rarely seen)
- Cell Attachment
- Carbohydrates also form the **glycocalyx** coat, the layer outside of the plasma membrane.

Functions of the glycocalyx:

- **Protection:** Cushions the plasma membrane and protects it from chemical injury
- **Immunity to infection:** Enables the immune system to recognize and selectively attack foreign organisms
- **Transplant compatibility:** Forms the basis for compatibility of blood transfusions, tissue grafts, and organ transplants
- **Cell adhesion:** Binds cells together so that tissues do not fall apart
- **Fertilization:** Enables sperm to recognize and bind to eggs
- **Embryonic development:** Guides embryonic cells to their destinations in the body

(ii) Explain why it is necessary to form a callus in order to grow genetically modified rice.

- Callus divide by mitosis to ensure all the genetically identical daughter cells would have the gene;
- Callus can be genetically modified by *A. tumefaciens*;
- Callus can be sub-cultured to increase the number of callus;
- Callus can be manipulated by auxin and cytokinin to induce root and shoot formation to form a plantlet;

[3]

(Any three)

(d) Edible vaccines have been made using other plants like tomato and banana. Suggest the advantages of using rice over other plants. [2]

- Rice can be stored for a long time;
- Rice forms a staple diet among many Asians
- Rice is versatile enough to be eaten with other dishes;

[Total: 13]

4 Planning question

DNA is the genetic material in all living organisms. DNA from an external source can be taken up by bacteria by a process known as transformation.

Plan an experiment to investigate the effect of varying plasmid concentration on the efficiency of *E. coli* transformation.

You may, or otherwise, also use the following equipment and materials:

- Solution of plasmid containing ampicillin resistance gene of concentration 100 Unit/ml
- Suspension of competent *E. coli* cells
- Distilled water
- LB broth (nutrient solution)
- Petri dishes containing LB agar medium and ampicillin (LB/amp plates)
- Sterilised microfuge tubes
- Micropipettes
- Bleach solution
- alcohol

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms

[Total:12]

Mark Scheme

1. Theoretical consideration / rationale of plan to justify the practical procedure [1]
2. Correct use of technical and scientific terms – any 2 of the following [1]

Relate theory to prac

Examples:

- DNA is the genetic material in all living organisms. Transformation is the process by which DNA from an external source is taken up by bacteria.
- The plasmid contains an ampicillin-resistance gene that confers resistance to the antibiotic.
- Bacteria that have taken up this plasmid are able to grow on LB/amp medium.
- Amp resistance gene serves as genetic marker to identify transformed cell
- When made competent by heat-shock, pores appear transiently in the membrane of bacteria. External DNA/plasmids may then enter the cell via these pores
- Consequently, as plasmid concentration increases, the efficiency of *E.coli* transformation will also increase proportionally.

3. An explanation of the dependent and independent variables involved [1]

- Independent variable: Concentration of plasmid (unit/ml)
- Dependent variable: Efficiency of *E. coli* transformation is represented by Number of colonies on LB/Amp plates

[A description of method used including the scientific reasoning behind the method]

4. Description of how the plasmid solution is prepared [1]

- i. Using plasmid solution of concentration 100 Unit/ml, make dilutions to obtain 5 different concentrations.

Concentration of plasmid (unit/ml)*	Volume of plasmid solution of conc 100 Units/ml (ml)*	Volume of water (ml)*
100	1.0	0.0
80	0.8	0.2
60	0.6	0.4
40	0.4	0.6
20	0.2	0.8
0	0	1.0

* or any other reasonable range / volume

OR

serial dilutions:

Concentration of plasmid (unit/ml)	Volume of stock (ml)	Volume of water (ml)
Undiluted	1.0	0.0
1 to 10 or 10^{-1}	0.1 of non dilute	0.9
1 to 100 or 10^{-2}	0.1 of 10^{-1}	0.9
1 to 1000 or 10^{-3}	0.1 of 10^{-2}	0.9
1 to 10000 or 10^{-4}	0.1 of 10^{-3}	0.9

5. Controlling factors that affect experiment (eg. Duration, temp, volumes,etc) [1]

6. Description of expt [1]

- Mix a fixed volume of the *E. coli* culture with a fixed volume (eg 10 μ l) of the each plasmid solution
- Heat-shock the bacteria-plasmid mixture
- Plate each sample on LB/amp plate.
- Seal the plates with tape and incubate for 24 hours in a 37°C (range 28-37) incubator oven.
- After an overnight incubation, count the number of colonies in each plate.

[How you will record your results and ensure they are as accurate and reliable as possible]

7. Control [1]

- For the control tube, replace plasmid solution with same volume of 10 μ l of distilled water/buffer solution.

Rationale:

To show that the presence of bacterial colonies on agar plates containing ampicillin is due to transformed bacteria which have taken up plasmid with amp resistance gene OR
in the absence of plasmid which carries the ampicillin resistant genes, there are no ampicillin resistant bacterial cells.

8. Accuracy and reliability [1]

- For each plasmid concentration, perform two replicates / repeat steps ___ to ___ twice
- Calculate average
- Check for at least 5 different concentrations of plasmid solutions

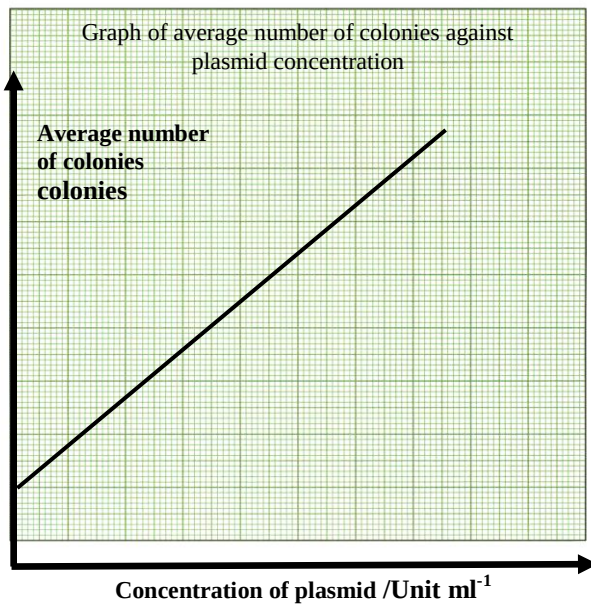
[Propose layout of results table and graph with clear headings and labels]

9. Table with appropriate headings [1]

Concentration of plasmid solution (unit/ml)	Number of colonies in each plate			
	Reading 1	Reading 2	Reading 3	Average

10. Graph [1]

- Clear label of axes with units and consistent headings of table
- propose expected trend

**11. Risk Assessment [1] – any 2**

- *E coli* is a bacteria and can cause disease.

Safety Precautions:

- Avoid contact - wear gloves and safety goggles.
- If in contact, wash off with water and antibacterial soap immediately.
- If skin is broken, obtain medical attention.
- Dispose of used microfuge tubes / pipette tips / inoculation loops in bleach solution
- Sterilize bench with alcohol after expt

12. Suitable diagram illustrating plates with bacteria / process of heat shock [1]

Free-response question Write your answers to this question on the separate answer paper provided. Your answer: • should be illustrated by large, clearly labelled diagrams, where appropriate; • must be in continuous prose, where appropriate; • must be set out in sections (a), (b) etc., as indicated in the question.		
5	(a) Rubisco is an important enzyme in the Calvin cycle. Outline how it might be possible to produce genetically modified plants with increased rubisco concentrations, or rubisco with enhanced efficiency. • Obtain an explant (from e.g. shoot tip); • Culture in a medium with (macro) and (micro) nutrients, sucrose as an energy source, and intermediate ratio of auxin to cytokinin. • Reference to aseptic conditions, e.g. surface sterilization. • Callus forms and divides by mitosis to form a mass of undifferentiated cells. • Isolate DNA containing rubisco that has a high efficiency from another plant species. • Use same restriction enzyme to cut DNA containing gene for rubisco and (Ti) plasmid; • Allow complementary sticky ends anneal, and DNA ligase to seal the nick by forming phosphodiester bond; • Recombinant DNA transformed into <i>Agrobacterium tumefaciens</i> bacteria cell by calcium chloride heat shock method; • Infect callus with transformed <i>Agrobacterium tumefaciens</i> bacteria which will integrate gene for rubisco into plant chromosome; • Subculture to increase number of callus with desired gene; • Increase cytokinin to auxin ration for shoot formation. • Increase auxin to cytokinin ratio for root formation. • Acclimatize plantlets in greenhouse; • Transfer to sterile soil outside; • Select for plants that have increased rubisco activity by measuring rate of photosynthesis; • Repeat whole process of plant tissue culture; (Half mark each)	[6]
	(b) Explain the basis of genetic engineering and how genetic engineering can improve the yield of one named crop plant and one named animal in solving the demand for food in the world. Basis (2 marks) Any two: • The genetic code is universal / will code for same amino acids in all organisms; • Central Dogma of molecular biology applies, • when gene from another species can still be expressed in the new host; • By transcription and translation for protein synthesis; • Nucleic acid is the basis of all life-forms; 3 marks for plant example and elaboration: A named crop: <i>Bt</i> corn	[8]

	The problem: The European corn borer is a pest that feeds on corn, hence reducing the amount of corn being sold. Nature of the infestation makes it difficult to deal with despite using chemical insecticides and biological control methods. Source of the gene: <i>Bacillus thuringiensis</i> (<i>Bt</i>) is a soil bacterium that encodes a <i>cry</i> protein (<i>Bt</i> toxin). Gene product in the crop: <i>Bt</i> toxin The solution: Transgenic corn plants containing the <i>Bt</i> gene will produce the protoxin which when ingested by the insect, is cleaved in the insect gut and the active <i>Bt</i> toxin causes gut cells to lyse, leading to the death of the insect. OR A named crop: <i>Flavr Savr</i> tomatoes The problem: Polygalacturonase (PG) degrades pectin in the tomato cell wall, causing tomatoes to soften, and hence a short shelf life. Source of the gene: Antisense gene Gene product in the crop: Antisense mRNA The solution: Antisense mRNA binds by complementary base-pairing to mRNA, preventing translation and hence reducing PG. This increases the shelf-life of tomato. OR Reject A named crop: Golden rice The problem: Childhood blindness in developing countries due to vitamin A deficiency in the diet, which comprised of rice primarily. Outer coat of the dehusked grains (aleurone layer) does not contain beta-carotene. Source of the gene: Two beta-carotene biosynthesis genes/ phytoene synthase (<i>psy</i>) gene and phytoene desaturase (<i>crt 1</i>) gene from the daffodil plant and soil bacterium, <i>Erwinia uredovora</i>. Gene product in the crop: beta-carotene produced 23 times more. The solution: Prevent vitamin A deficiency as beta-carotene is a precursor of Vitamin A. AND 3 marks for animal example and elaboration: A named crop: GM salmon The problem: Atlantic salmon can only grow in summer because their growth hormone gene is switched off in winter. Source of the gene: Antifreeze protein (AFP) promoter from ocean pout to drive expression of Chinook salmon growth hormone gene. Gene product in the crop: Cold conditions stimulate transcription of growth hormone gene. The solution: GM salmon are larger and reach sexual maturity sooner than wild salmon. They inhibit increased feed use efficiency so they convert more food into tissues more rapidly.	
(c)	Discuss how the human genome project can help parents customise their babies and the possible implications of doing so. Info from HGP [1] must have both to gain 2 marks. - One can use the information from Human Genome Project to identify <u>a wide range /genome-wide range (or state number of genes);</u> - of disease causing alleles, alleles that give desirable traits such as height, intelligence, strength endurance etc; (Important to state that it is a genome-wide range, since prior to HGP, other genetic traits would have been known). How info use to create Designer Babies [1] either one is a necessary point. - Based on info from HGP, Designer babies could 1) be selected for using Pre-	[6]

		implantation Genetic Diagnosis (PGD)/(OWTTE) that are disease-free or have desirable traits; • Or based on the HGP, designer babies could 2) go through genetic modification for enhancements in desirable traits/removal of undesirable traits discovered through info from HGP; Possible Benefits (at least 2 points to gain full marks) • Babies that are born would not have to suffer from genetic defects as those that have disease-causing alleles are not implanted or have their genes modified to correct the disease; • Designer siblings of children who have genetic diseases may be created to provide matching and healthy stem cells for therapy; • Determine the sex of the child as this might prevent babies who have diseases like haemophilia and Duchenne Muscular Dystrophy, Fragile X syndrome, only show themselves in male babies; Ethical concerns (at least 2 points to gain full marks) • For the selection of embryos to implant in PGD, those that are not selected are discarded or unused which will raise ethical concerns since these embryos have been fertilized. • Designer siblings may be feel that their existence was for a cure rather than parents wanting a child/feel inferior to the child who is sick • Choosing the sex of babies can raise ethical concerns especially in societies where boys are valued more highly than girls, tipping the sex-ratio. • Designer babies with genetical enhancements (cf gene therapy) to add genes for HIV resistance or longevity or a high IQ? • May lead to a social divide between those who can afford “designer babies”/designer genome and those that cannot.	
			[Total: 20]